

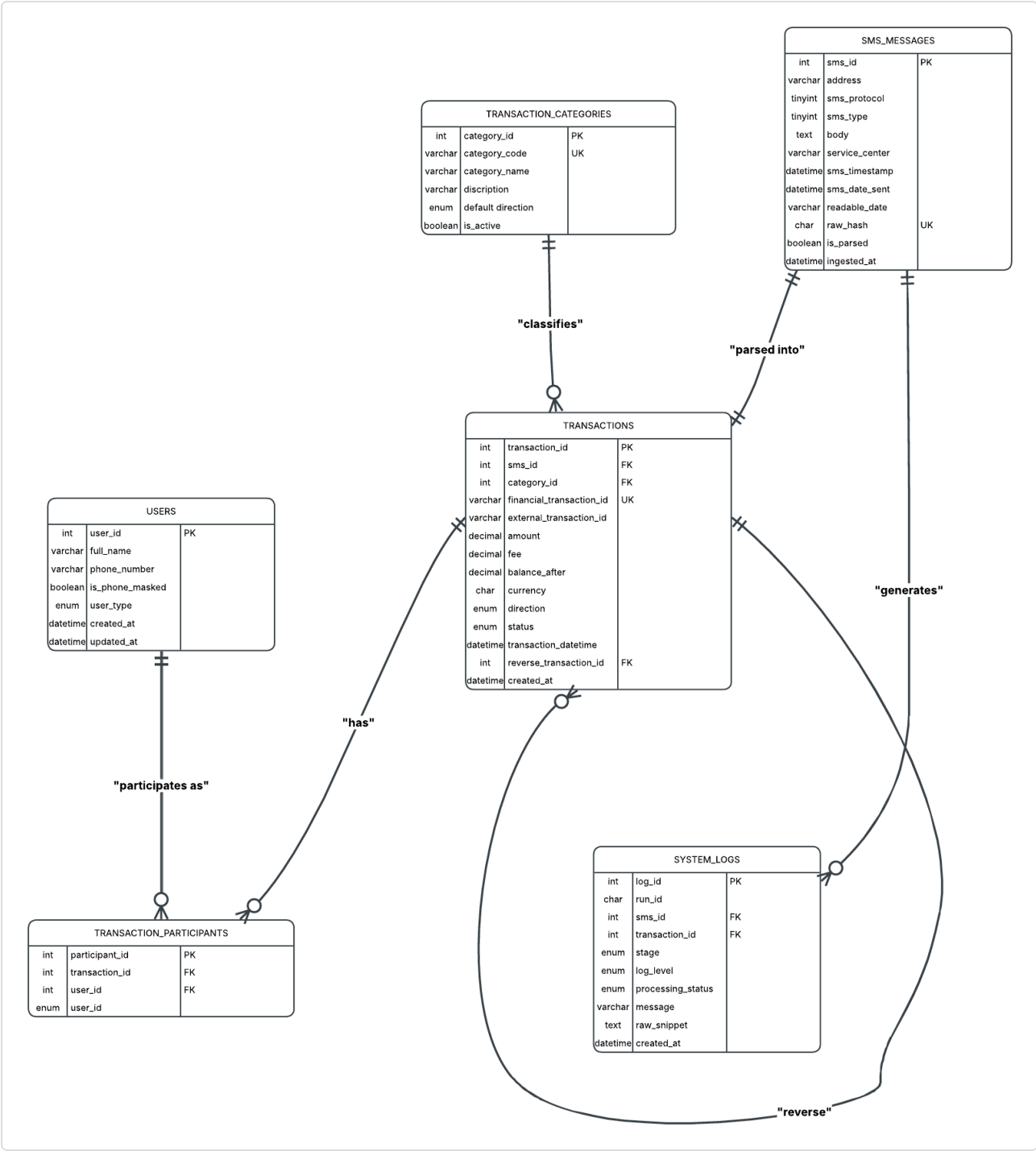
Database Design Document - MoMo SMS Data Processing System

Database Design Document — MoMo SMS Data Processing System

Week 2 deliverable · Urban Mobility Data Explorer (MoMo ETL) team

- ERD source (editable): [erd_diagram.drawio](#) — open at app.diagrams.net
 - ERD image: [erd_diagram.png](#)
 - SQL implementation: [../database/database_setup.sql](#)
 - JSON serialization examples: [../examples/json_schemas.json](#)
-

1. Entity Relationship Diagram



ERD

Six entities: users , transaction_categories , sms_messages , transactions , transaction_participants , system_logs .

Relationship	Cardinality	Notes
transaction_categories → transactions	1:M	Every transaction has exactly one category
sms_messages → transactions	1:0..1	A message parses into at most one transaction (sms_id is UNIQUE on

Relationship	Cardinality	Notes
		transactions)
transactions → transaction_participants	1:M	A transaction has 1-3 participant rows
users → transaction_participants	1:M	A user participates in many transactions over time
users ↔ transactions	M:N, resolved by transaction_participants	See §2
transactions → transactions (self)	1:0..1	reverses_transaction_id , set only when status = REVERSED
sms_messages / transactions → system_logs	1:M	ETL audit trail, both FKs nullable

2. Design Rationale (why the schema looks like this)

The schema was derived directly from `data/raw/momo.xml` , not from a generic template — every design decision below traces back to a specific message pattern found in that file.

Why a junction table instead of `sender_id` / `receiver_id` columns. The eleven distinct SMS shapes in the data don't all describe a simple two-party transfer. A bank deposit names no counterparty at all; a merchant payment names one; a peer transfer names one; and an **agent withdrawal** (“*You Account Holder... have via agent: Agent John (...), withdrawn...*”) names two — the account holder and the agent — for a single transaction. Two fixed FK columns can't represent that variability cleanly, so `transaction_participants` resolves `users` ↔ `transactions` as a real many-to-many relationship, with a `role` (`SENDER` / `RECEIVER` / `AGENT` / `ACCOUNT_HOLDER`) on each edge. This is the required M:N relationship, and it is evidenced by the source data rather than added just to satisfy the rubric.

Why `phone_number` is nullable and not unique. The same real-world contact appears both masked (`*****013`) and in full (`250791666666`) across different messages, and bank deposits carry only a settlement account number, not a phone number. A `UNIQUE` constraint on phone would break the first time a masked and unmasked number collided or a deposit arrived with no number at all, so `users` uses a surrogate `user_id` and treats phone as an informational, best-effort field.

Why `status` and the self-referencing `reverses_transaction_id`. The corpus contains not just completed transactions but explicit failures (“*...has failed at...*”) and reversals (“*A reversal has been initiated...*”). Modeling only “happy path” transactions would silently drop real data the ETL pipeline has to handle, so `status` is a first-class `ENUM` and reversals link back to the original transaction via a nullable self-FK rather than a separate table.

Why `sms_messages` is a separate table from `transactions`. Keeping the raw SMS verbatim (1:1 with its parsed transaction via a `UNIQUE` FK) preserves an audit trail back to the original text and lets `system_logs` record processing outcomes — including dead-letter entries — for messages that never produce a transaction at all (`transaction_id` is nullable there for exactly that reason).

Why `financial_transaction_id` is `UNIQUE` but nullable. It is MoMo's own transaction identifier (`TxId` / “Financial Transaction Id” in the SMS body) and doubles as the ETL's idempotency key — re-ingesting the same backup file must not create duplicate rows. It's nullable because a handful of message types (bank deposits, mobile transfers) don't surface one.

3. Data Dictionary

users

Column	Type	Constraints	Description
user_id	INT	PK, AUTO_INCREMENT	Surrogate key
full_name	VARCHAR(100)	NOT NULL	Name as it appears in the SMS body
phone_number	VARCHAR(15)	NULL	May be masked (*****013), full, or absent
is_phone_masked	BOOLEAN	NOT NULL, DEFAULT FALSE	True when phone_number contains * characters
user_type	ENUM	NOT NULL, DEFAULT 'CUSTOMER'	CUSTOMER / AGENT / MERCHANT / SYSTEM
created_at, updated_at	DATETIME	NOT NULL	Row bookkeeping

transaction_categories

Column	Type	Constraints	Description
category_id	INT	PK, AUTO_INCREMENT	Surrogate key
category_code	VARCHAR(30)	NOT NULL, UNIQUE	Stable key used by etl/categorize.py
category_name	VARCHAR(100)	NOT NULL	Dashboard label
description	VARCHAR(255)	NULL	
default_direction	ENUM('CREDIT','DEBIT')	NOT NULL	Typical balance effect

sms_messages

Column	Type	Constraints	Description
sms_id	INT	PK, AUTO_INCREMENT	Surrogate key
address	VARCHAR(30)	NOT NULL	e.g. "M-Money"
body	TEXT	NOT NULL	Raw, unparsed SMS text
sms_timestamp	DATETIME	NOT NULL	Parsed from the date epoch-ms attribute
raw_hash	CHAR(64)	NOT NULL, UNIQUE	SHA-256 of the raw element; dedupes re-ingested backups
is_parsed	BOOLEAN	NOT NULL, DEFAULT FALSE	Set once a transactions row exists
service_center, readable_date, sms_protocol, sms_type, sms_date_sent, ingested_at	—	NULL	Secondary/debug fields carried from the XML

transactions

Column	Type	Constraints	Description
transaction_id	INT	PK, AUTO_INCREMENT	Surrogate key
sms_id	INT	FK → sms_messages, UNIQUE, NOT NULL	1:1 provenance link
category_id	INT	FK → transaction_categories, NOT NULL	
financial_transaction_id	VARCHAR(30)	UNIQUE, NULL	MoMo's own TxId; ETL idempotency key
external_transaction_id	VARCHAR(30)	NULL	Present on some third-party debits
amount	DECIMAL(12,2)	NOT NULL, CHECK (amount > 0)	
fee	DECIMAL(12,2)	NOT NULL, DEFAULT 0, CHECK (fee >= 0)	
balance_after	DECIMAL(12,2)	NULL, CHECK (balance_after >= 0)	Not every message reports it
currency	CHAR(3)	NOT NULL, DEFAULT 'RWF'	
direction	ENUM('CREDIT','DEBIT')	NOT NULL	
status	ENUM('COMPLETED','FAILED','PENDING','REVERSED')	NOT NULL, DEFAULT 'COMPLETED'	
transaction_datetime	DATETIME	NOT NULL	Timestamp embedded in the SMS text
reverses_transaction_id	INT	FK → transactions (self), NULL	Set only when status = REVERSED

transaction_participants (junction table)

Column	Type	Constraints	Description
participant_id	INT	PK, AUTO_INCREMENT	Surrogate key
transaction_id	INT	FK → transactions, NOT NULL	
user_id	INT	FK → users, NOT NULL	
role	ENUM('SENDER','RECEIVER','AGENT','ACCOUNT HOLDER')	NOT NULL	
—	—	UNIQUE (transaction_id, role)	A transaction has at most one of each role

system_logs

Column	Type	Constraints	Description
log_id	INT	PK, AUTO_INCREMENT	Surrogate key
sms_id	INT	FK → sms_messages, NULL, ON DELETE SET NULL	
transaction_id	INT	FK → transactions, NULL, ON DELETE SET NULL	
stage	ENUM('PARSE','CLEAN','CATEGORIZE','LOAD')	NOT NULL	
log_level	ENUM('INFO','WARNING','ERROR')	NOT NULL, DEFAULT 'INFO'	
processing_status	ENUM('SUCCESS','FAILED','SKIPPED')	NOT NULL	
message	VARCHAR(500)	NOT NULL	
raw_snippet	TEXT	NULL	Populated for dead-letter entries

4. JSON Serialization Mapping

`examples/json_schemas.json` gives one example object per table plus two composite shapes:

- `transaction_flat` — a bare `transactions` row, FK columns only (what `SELECT *` looks like serialized).
- `transaction_detailed` — the **complex nested object** the assignment asks for: a full transaction with its `category` resolved and every `transaction_participants` row expanded into a nested `user` object. This is the shape a `GET /transactions/{id}` endpoint (see `api/schemas.py`) or `web/chart_handler.js` would actually consume.
- `agent_withdrawal_example` / `reversal_example` — show the 3-party M:N case and the self-referencing reversal in JSON form.
- `dashboard_analytics_example` — an aggregate shape (`GROUP BY category`) for chart data.

5. Verified on MySQL 8.0 (db-fiddle.com, MySQL v8)

The full `database/database_setup.sql` (all 6 `CREATE TABLE` statements, all sample `INSERT` s) was run unmodified against a real MySQL 8.0 instance, not just validated locally, because `CHECK` constraints are silently ignored before MySQL 8.0.16 — a schema that *parses* is not proof it *enforces* anything. Screenshots below are from that run.

5.1 Sample query — joins + the M:N junction resolved correctly

```
SELECT t.transaction_id, t.financial_transaction_id, t.amount, t.direction, t.status,
       tc.category_name,
       GROUP_CONCAT(CONCAT(tp.role, ':', u.full_name) ORDER BY tp.role SEPARATOR ' | ') AS
participants
FROM transactions t
JOIN transaction_categories tc ON tc.category_id = t.category_id
JOIN transaction_participants tp ON tp.transaction_id = t.transaction_id
JOIN users u ON u.user_id = tp.user_id
GROUP BY t.transaction_id
ORDER BY t.transaction_datetime;
```

Result includes, e.g., transaction 9: AGENT:Agent John | ACCOUNT_HOLDER:Account Holder — a single transaction correctly resolved to two distinct users in two distinct roles, proving the junction table works.

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Schema SQL

```
1 -- =====
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6 -- enforcement;      on 5.7/8.0.0-8.0.15 CHECK clauses parse but are
7 -- ignored)
8 -- How to run      : mysql -u <user> -p < database_setup.sql
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13 -- People or entities that appear as a party (sender/receiver/agent)
14 -- in a transaction. Populated from names + phone numbers found in
15 -- the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 -- Full transaction detail joined with category and every participant
2 -- (proves the M:N junction resolves correctly)
3 SELECT t.transaction_id, t.financial_transaction_id, t.amount, t.direction,
4        t.status,
5        tc.category_name,
6        GROUP_CONCAT(CONCAT(tp.role, ':', u.full_name) ORDER BY tp.role
7        SEPARATOR ' | ') AS participants
8 FROM transactions t
9 JOIN transaction_categories tc ON tc.category_id = t.category_id
10 JOIN transaction_participants tp ON tp.transaction_id = t.transaction_id
11 JOIN users u ON u.user_id = tp.user_id
12 GROUP BY t.transaction_id
13 ORDER BY t.transaction_datetime;
```

Results

Copy as Markdown

2	73214484437	1000.0	DEBIT	COMPLETED	Payment to Code Holder	SENDER:Account Holder RECEIVER:Jane Smith
3	null	40000.0	CREDIT	COMPLETED	Bank Deposit	RECEIVER:Account Holder
4	null	10000.0	DEBIT	COMPLETED	Transfer to Mobile Number	SENDER:Account Holder RECEIVER:Samuel Carter
5	13913173274	2000.0	DEBIT	COMPLETED	Airtime Bill Payment	SENDER:Account Holder
6	14264876273	2000.0	DEBIT	COMPLETED	Cash Power Bill Payment	SENDER:Account Holder
7	14324965479	2000.0	DEBIT	COMPLETED	Bundles and Packs Purchase	SENDER:Account Holder
8	13947831685	25000.0	DEBIT	COMPLETED	Third Party Transaction	SENDER:Account Holder RECEIVER:DIRECT PAYMENT LTD
9	98765432101	15000.0	DEBIT	COMPLETED	Withdrawal from Agent	AGENT:Agent John ACCOUNT_HOLDER:Account Holder

Join query result on MySQL 8.0

5.2 Basic CRUD operations

Operation	Statement	Verified result
CREATE	INSERT INTO users (...) VALUES ('Grace Uwimana', ...)	grace_uwimana_created = 1
UPDATE	UPDATE transactions SET amount = 1500.00 WHERE transaction_id = 2	txn2_amount_after_update = 1500.0
DELETE	DELETE FROM system_logs WHERE stage='PARSE' AND processing_status='FAILED'	dead_letter_logs_remaining = 0

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14 -- the SMS body text.
15 --
16 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
17 --

```

Text to DDL

Query SQL

```

1 -- Basic CRUD demonstration (Schema SQL panel re-seeds the DB before each
2 -- Run,
3 -- so CREATE/UPDATE/DELETE + verification READ must happen together in one
4 -- query batch)
5 --
6 -- CREATE
7 -- INSERT INTO users (full_name, phone_number, user_type) VALUES ('Grace
8 -- Uwimana', '250788123456', 'CUSTOMER');
9 --
10 -- UPDATE
11 -- UPDATE transactions SET amount = 1500.00 WHERE transaction_id = 2;
12 --
13 -- DELETE
14 -- DELETE FROM system_logs WHERE stage = 'PARSE' AND processing_status =
15 -- 'FAILED';
16 --
17 -- READ: verify all three writes landed
18 --
19 -- SELECT
20 -- (SELECT COUNT(*) FROM users WHERE full_name = 'Grace Uwimana')

```

Copy as Markdown

Results

There are no results to be displayed.

Query #2 Execution time: 0.62ms

There are no results to be displayed.

Query #3 Execution time: 0.56ms

There are no results to be displayed.

Query #4 Execution time: 0.38ms

grace_uwimana_created	txn2_amount_after_update	dead_letter_logs_remaining
1	1500.0	0

CRUD operations verified in one batch

5.3 Additional sample query — top senders by total amount

```

SELECT u.full_name, COUNT(*) AS transactions_sent, SUM(t.amount) AS total_sent
FROM transaction_participants tp
JOIN users u ON u.user_id = tp.user_id
JOIN transactions t ON t.transaction_id = tp.transaction_id
WHERE tp.role = 'SENDER' AND t.status = 'COMPLETED'
GROUP BY u.full_name
ORDER BY total_sent DESC;

```

Database: MySQL v8

Run Save Load Example Collaborate

Sign in Have any feedback?

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Schema SQL

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8 -- but are ignored)
9 --
10 -- How to run : mysql -u <user> -p < database_setup.sql
11 -- =====

```

Text to DDL

Query SQL

```

1 SELECT u.full_name, COUNT(*) AS transactions_sent,
2 SUM(t.amount) AS total_sent
3 FROM transaction_participants tp
4 JOIN users u ON u.user_id = tp.user_id
5 JOIN transactions t ON t.transaction_id = tp.transaction_id
6 WHERE tp.role = 'SENDER' AND t.status = 'COMPLETED'
7 GROUP BY u.full_name
8 ORDER BY total_sent DESC;

```

Copy as Markdown

Results

Query #1 Execution time: 1.93ms

full_name	transactions_sent	total_sent
Account Holder	6	42000.0
Jane Smith	1	2000.0

Top senders query result

6. Unique Rules Enforced (security & accuracy)

Each rule below was actually **triggered against MySQL 8.0** (not just declared) — the screenshots below show the real rejection, confirming CHECK constraints are active on this engine version (they are silently ignored before MySQL 8.0.16).

#	Rule	Purpose	Observed error
1	UNIQUE (financial_transaction_id) on transactions	ETL idempotency — re-ingesting the same MoMo SMS backup must never duplicate a transaction	Duplicate entry '76662021700' for key 'transactions.financial_transaction_id'
2	CHECK (amount > 0)	An amount ≤ 0 signals a parsing bug upstream, not a real transaction	Check constraint 'chk_txn_amount_positive' is violated.
3	FOREIGN KEY (category_id) ... ON DELETE RESTRICT	Every transaction must resolve to a known category; blocks orphan inserts	Cannot add or update a child row: a foreign key constraint fails (...fk_txn_category...)
4	ON DELETE RESTRICT from transactions → transaction_categories	A category still referenced by transaction history cannot be deleted, protecting historical classification	Cannot delete or update a parent row: a foreign key constraint fails (...fk_txn_category...)
5	UNIQUE (transaction_id, role) on transaction_participants	A transaction cannot have two SENDERS, two AGENTS, etc.	Duplicate entry '1-SENDER' for key 'transaction_participants.uq_txn_role'
6	CHECK (fee >= 0)	Rejects an impossible negative fee from a bad parse	Check constraint 'chk_txn_fee_nonnegative' is violated.

CHECK (balance_after IS NULL OR balance_after >= 0) uses the identical chk_* mechanism verified in rules 2 and 6 above, so one working example of each mechanism (UNIQUE, CHECK, FK RESTRICT) is treated as sufficient proof it applies engine-wide; it was not screenshotted separately.

6.1 Screenshot evidence

Rule 1 — UNIQUE (financial_transaction_id)

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14 -- in a transaction. Populated from names + phone numbers found in
15 -- the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 |- UNIQUE constraint demo: financial_transaction_id is the MoMo idempotency
2 -- key
3 -- (the ETL upserts on this value so a re-ingested SMS never duplicates a
4 -- transaction).
5 -- Uses a fresh sms_id (12) so only the financial_transaction_id UNIQUE is
6 -- being tested.
7 INSERT INTO sms_messages (sms_id, address, body, sms_timestamp, raw_hash)
8 VALUES (12, 'M-Money', 'duplicate test SMS', '2024-06-01 00:00:00',
9 SHA2('dup-test', 256));
10
11 INSERT INTO transactions (sms_id, category_id, financial_transaction_id,
12 amount, direction, transaction_datetime)
13 VALUES (12, 1, '76662021700', 2000.00, 'CREDIT', '2024-06-01 00:00:00');
```

Results

Query Error: Duplicate entry '76662021700' for key 'transactions.financial_transaction_id'

Rule 2 — CHECK (amount > 0)

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Text to DDL

Query SQL

```
1 |- CHECK constraint demo: amount must be strictly positive.
2 -- A negative/zero amount here would signal a parsing bug upstream in
3 -- etl/clean_normalize.py,
4 -- not a valid MoMo transaction, so the database rejects it outright.
5 INSERT INTO transactions (sms_id, category_id, amount, direction,
6 transaction_datetime)
7 VALUES (12, 1, -500.00, 'DEBIT', '2024-06-01 00:00:00');
```

Results

Query Error: Check constraint 'chk_txn_amount_positive' is violated.

Rule 3 — FOREIGN KEY (category_id)

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```

Text to DDL

Query SQL

```
1 -- FOREIGN KEY constraint demo: category_id must reference a real
2 -- transaction_categories row.
3 -- Guards against a bug in etl/categorize.py emitting an unrecognized
4 -- category code.
5 INSERT INTO sms_messages (sms_id, address, body, sms_timestamp, raw_hash)
6 VALUES (12, 'M-Money', 'fk test SMS', '2024-06-01 00:00:00', SHA2('fk-test',
7 256));
8
9 INSERT INTO transactions (sms_id, category_id, amount, direction,
10 transaction_datetime)
11 VALUES (12, 999, 500.00, 'DEBIT', '2024-06-01 00:00:00');
```

Results

Query Error: Cannot add or update a child row: a foreign key constraint fails ('test'.transactions', CONSTRAINT 'fk_txn_category' FOREIGN KEY ('category_id') REFERENCES 'transaction_categories' ('category_id') ON DELETE RESTRICT ON UPDATE CASCADE)

Rule 4 — ON DELETE RESTRICT

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16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 |- ON DELETE RESTRICT demo: a category still referenced by transactions
2 -- cannot be deleted.
3 -- Protects transaction history from silently losing its category
4 -- classification.
5 DELETE FROM transaction_categories WHERE category_id = 1;
```

Results

Query Error: Cannot delete or update a parent row: a foreign key constraint fails ('test'.transactions', CONSTRAINT 'fk_txn_category' FOREIGN KEY ('category_id') REFERENCES 'transaction_categories' ('category_id') ON DELETE RESTRICT ON UPDATE CASCADE)

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Conduct Online SQL Assessments in minutes

Schema

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Query Error:

Duplicate entry '1-SENDER' for key 'transaction_particip

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SQL Assessments
in minutes

Schema SQLQuery SQL

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Results

Query Error:
Check
constraint
'chk_txn_fee_nonne
is violated.

7. Mapping Back to the Repository

Deliverable	File
ERD (editable)	docs/erd_diagram.drawio
ERD (image)	docs/erd_diagram.png
SQL DDL + DML	database/database_setup.sql
JSON schemas	examples/json_schemas.json
This document	docs/database_design.md
Verification screenshots	docs/screenshots/01 – 08_*.png